

CHEROKEE COUNTY AP, TX, USA

WMO: 720298

Lat: 31.869N

Lon: 95.218W

Elev: 206

StdP: 98.87

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-1.2	-11.8	1.4	1.8	-9.5	1.7	3.9	8.5	14.1	7.9	14.2	2.5	0	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	36.3	22.9	35.2	23.1	34.0	23.3	25.8	30.5	25.5	30.3	25.1	29.9	2.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.4	27.1	24.3	19.7	26.7	23.9	19.3	26.6	80.6	30.4	79.0	30.3	77.5	30.1	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.6	5.6	-6.8	37.7	2.4	1.5	-8.6	38.8	-10.0	39.6	-11.4	40.5	-13.1	41.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	9.1	11.2	15.2	18.9	22.9	26.8	28.0	28.5	25.8	20.2	14.3	10.2
	DBStd	8.03	5.57	5.73	5.03	3.89	3.31	2.01	1.83	2.18	2.71	4.36	5.01	5.35
	HDD10.0	235	84	49	16	2	0	0	0	0	0	1	17	66
	HDD18.3	1089	288	206	121	39	7	0	0	0	1	33	137	258
	CDD10.0	3629	57	83	176	269	401	504	558	574	473	316	147	71
	CDD18.3	1441	2	7	22	57	149	254	299	316	224	90	17	4
	CDH23.3	13813	4	22	101	336	1147	2521	3258	3653	2090	616	58	8
	CDH26.7	6052	0	1	8	54	373	1115	1575	1869	889	163	5	0
Wind	WSAvg	2.7	2.9	3.1	3.2	3.3	2.8	2.4	2.2	2.2	2.0	2.4	2.7	2.8
Precipitation	PrecAvg	1082	81	80	98	106	116	96	66	59	98	97	90	96
	PrecMax	1599	195	152	267	307	269	251	163	159	253	276	184	218
	PrecMin	634	7	20	22	21	17	10	14	2	7	1	10	20
	PrecStd	226	52	32	66	72	65	75	36	40	64	76	46	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	26.0	27.7	30.3	33.0	36.2	37.4	38.0	36.0	32.0	27.5	24.3
		MCWB	15.5	16.6	18.2	19.9	22.3	22.0	22.7	21.8	21.5	23.1	20.6	18.7
	2%	DB	21.8	23.4	26.1	28.0	31.6	34.7	36.0	37.0	34.2	30.2	25.1	22.5
		MCWB	15.3	17.1	18.1	19.4	22.0	22.7	22.8	22.8	22.6	22.3	19.2	18.5
	5%	DB	19.8	21.5	24.2	26.9	30.2	33.0	34.8	35.6	32.8	28.5	23.0	20.8
		MCWB	15.6	16.9	17.4	19.0	21.9	23.3	23.4	23.6	22.7	21.3	18.5	18.0
	10%	DB	17.7	19.9	22.5	25.3	28.9	32.2	33.1	34.2	31.5	27.1	22.0	18.9
		MCWB	14.4	16.4	16.6	18.4	21.7	23.4	23.8	23.4	22.4	20.0	18.5	16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	20.3	20.8	23.2	24.7	26.2	26.1	26.3	26.0	24.8	22.4	21.1
		MCDB	20.7	22.8	23.9	26.1	28.8	30.8	31.1	30.9	30.0	28.1	24.9	22.3
	2%	WB	18.0	19.4	20.1	21.8	23.9	25.5	25.6	25.7	25.3	24.0	21.2	19.9
		MCDB	19.6	21.4	23.1	25.4	28.3	29.9	30.6	30.7	29.4	27.7	23.2	21.5
	5%	WB	17.2	18.3	19.2	21.0	23.3	25.1	25.2	25.4	24.7	23.3	20.1	18.7
		MCDB	18.8	20.4	22.2	24.3	27.6	29.4	30.2	30.6	28.7	26.6	22.3	20.5
	10%	WB	15.0	17.0	18.3	20.2	22.7	24.6	24.9	25.1	24.1	22.5	18.9	16.9
		MCDB	17.1	19.1	21.4	23.5	26.9	28.9	29.8	30.2	28.0	25.4	21.4	18.4
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.6	10.2	10.5	9.9	9.6	10.2	10.6	10.1	10.2	9.5	9.1
		MCDBR	10.9	10.5	10.8	11.1	11.0	11.0	12.0	12.1	11.4	10.7	10.1	10.5
		MCWBR	7.0	6.5	5.2	4.9	3.8	2.8	2.6	2.7	3.3	4.2	5.9	7.7
	5% WB	MCDBR	9.1	9.2	9.4	9.4	9.5	9.1	9.8	10.2	9.5	9.1	9.2	10.0
		MCWBR	7.5	6.9	6.3	5.4	4.0	3.1	2.9	2.7	3.1	4.4	6.9	8.5
Clear-Sky Solar Irradiance	taub	0.318	0.337	0.357	0.399	0.432	0.446	0.457	0.456	0.417	0.359	0.349	0.325	
	taud	2.518	2.474	2.425	2.313	2.271	2.276	2.281	2.278	2.340	2.482	2.463	2.508	
	Ebn at Noon	909	922	924	892	861	844	833	830	852	885	862	879	
	Edh at Noon	86	99	111	130	136	135	134	133	120	97	90	83	
All-Sky Solar Radiation	RadAvg	2.79	3.12	4.33	5.37	5.82	6.32	6.41	6.02	5.09	4.13	3.01	2.43	
	RadStd	0.26	0.54	0.41	0.35	0.43	0.52	0.46	0.45	0.46	0.57	0.37	0.31	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83

Nomenclature: See separate page